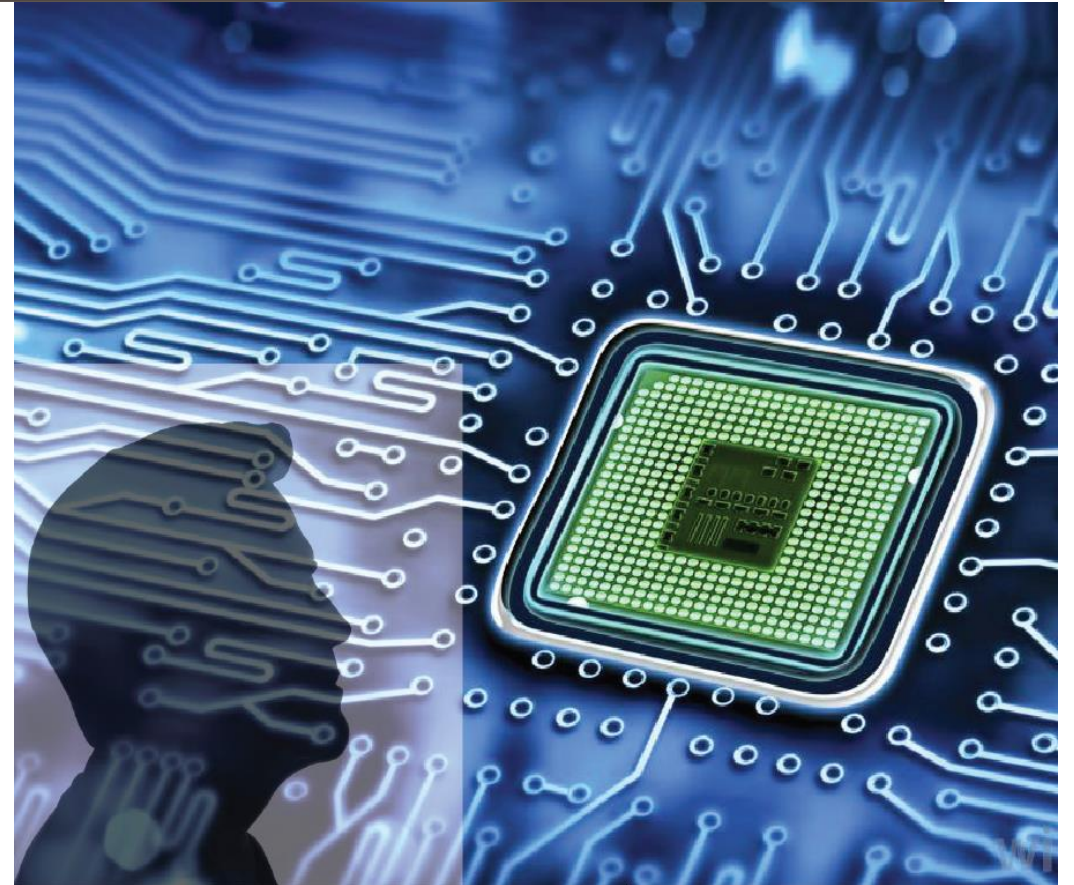


VLSI DESIGN FLOW

Week-2, Lecture-2 Introduction

Sneh Saurabh
21st August, 2024



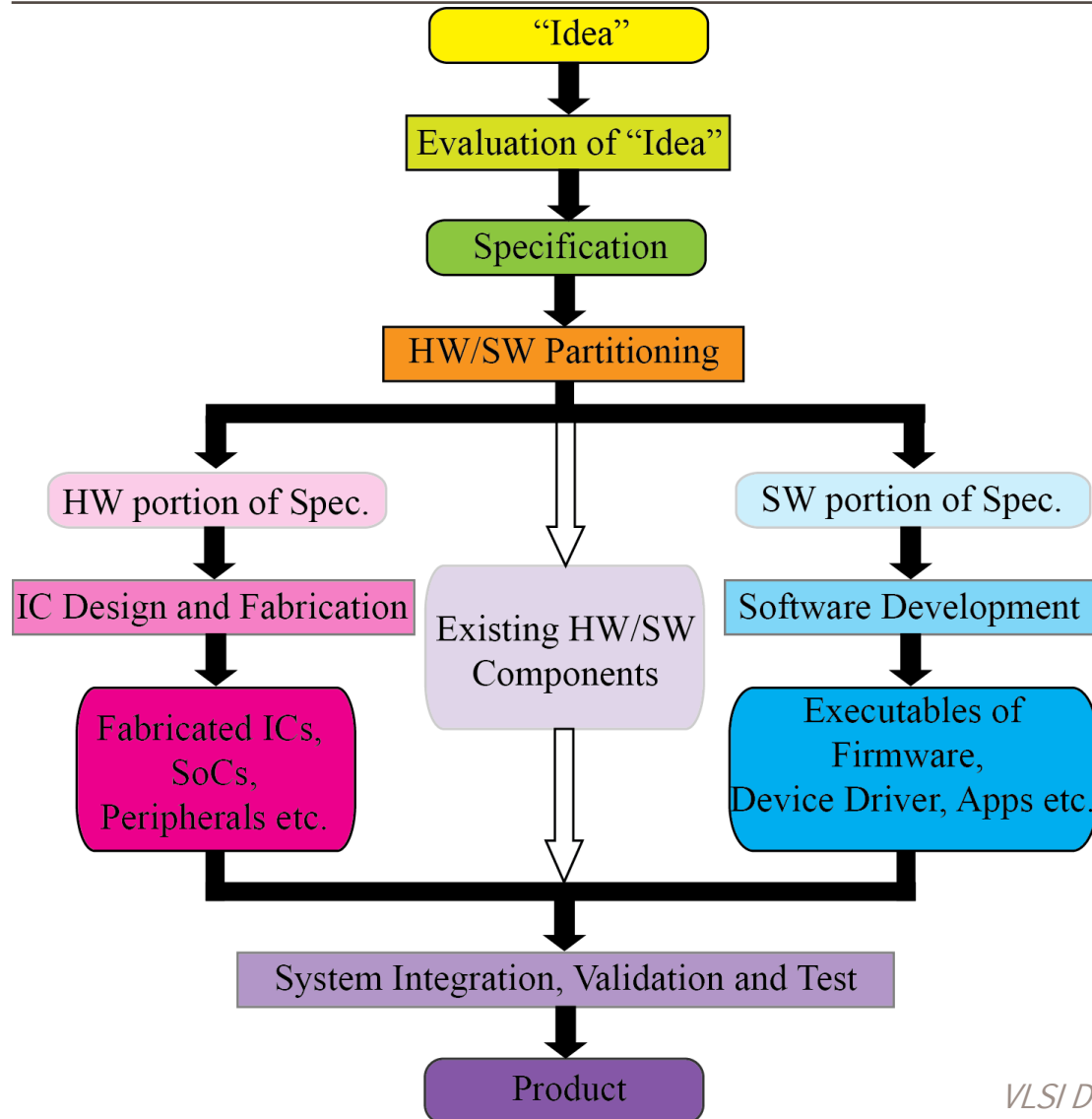
VDF: Revision/Announcements

- Project Topics



Pre-RTL Methodologies

System-level Design: Top View



Evaluation of "idea":

- Market requirement
- Financial viability
- Technical feasibility

Preparing specifications:

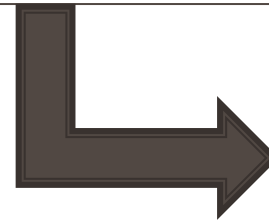
- Features (functionality)
- PPA
- Time to market (TTM)

HW—SW Partitioning:

- Identify components
- Determine which components to implement in HW/SW

- HW/SW Development (separately)
- System Integration, Validation, Test
- Final Product

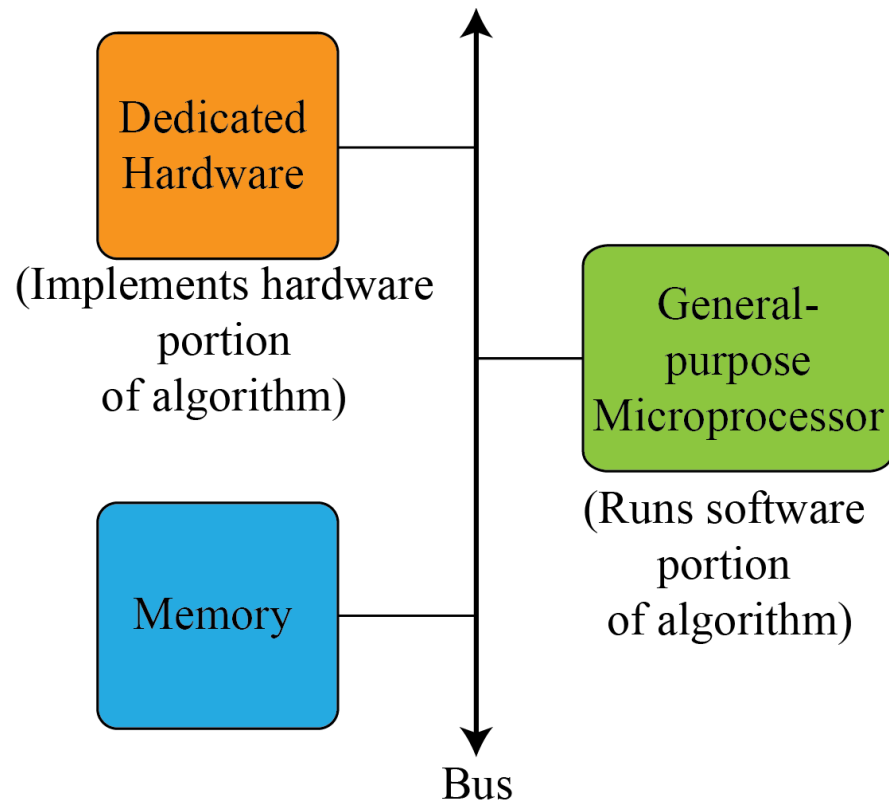
Pre-RTL Methodologies



Hardware—software
Partitioning

Hardware—Software Partitioning: Motivation

Motivation: Exploit the merits of both hardware and software by choosing right *combination of hardware and software* to implement a given function



- Hardware: usually runs as parallel circuits and can have very good PPA
 - Can be implemented in full custom IC, ASIC or FPGA
- Software: usually run sequentially on a general purpose processor

	Hardware	Software
Performance	High	Low
Cost	High	Low
Risk due to bug	High	Low
Customization	Low	High
Development Time	High	Low

Hardware/Software Partitioning: Example

Video Compression

- Algorithm can be divided in two main parts:
 1. Computing Discrete Cosine Transform (DCT):
Performed multiple times, bottleneck
 2. Frame Handling and other computation

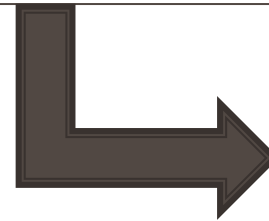


- **DCT**
 - Hardware for computing DCT
 - Can be computed using parallel circuits
 - Several orders of magnitude faster and more energy efficient implementation in hardware



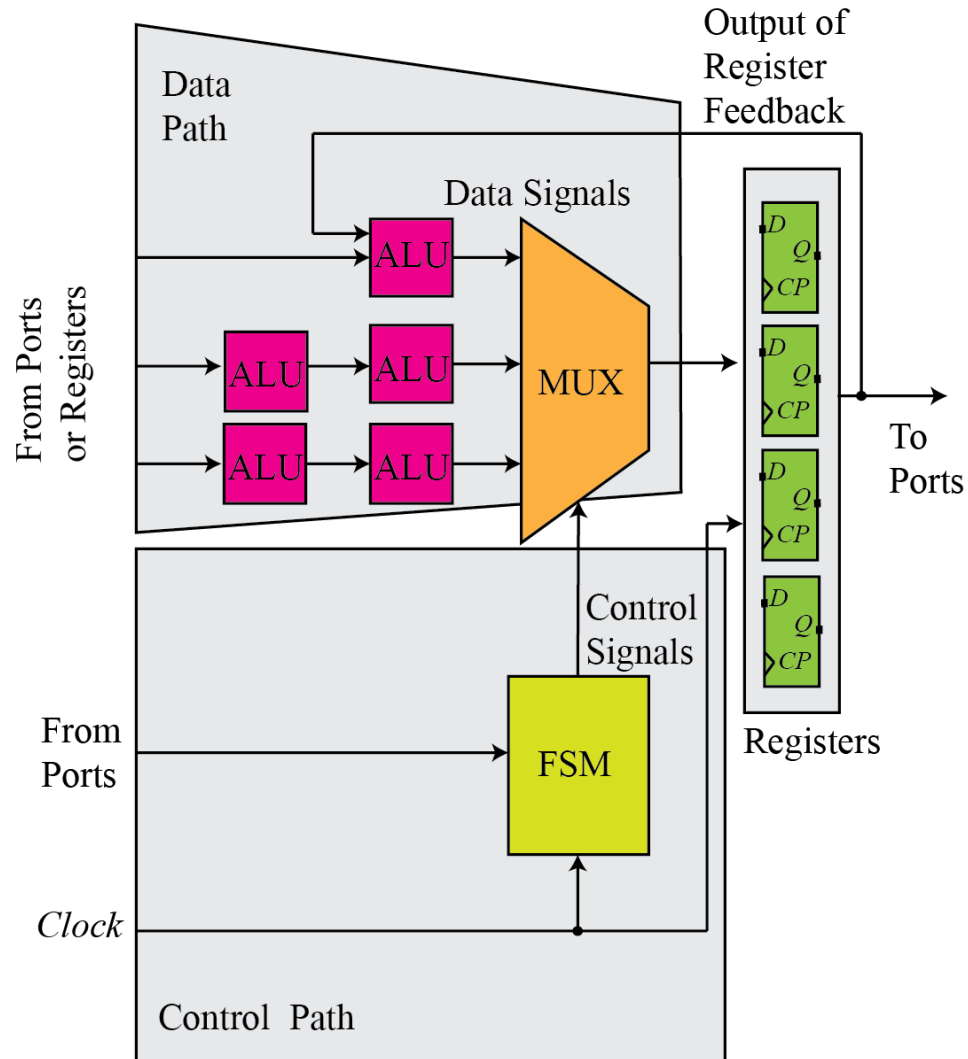
- **Frame Handling and other computation**
 - Software on a general purpose microprocessor
 - Provides Flexibility

Pre-RTL Methodologies



Functional
Specification to RTL

RTL: General Structure

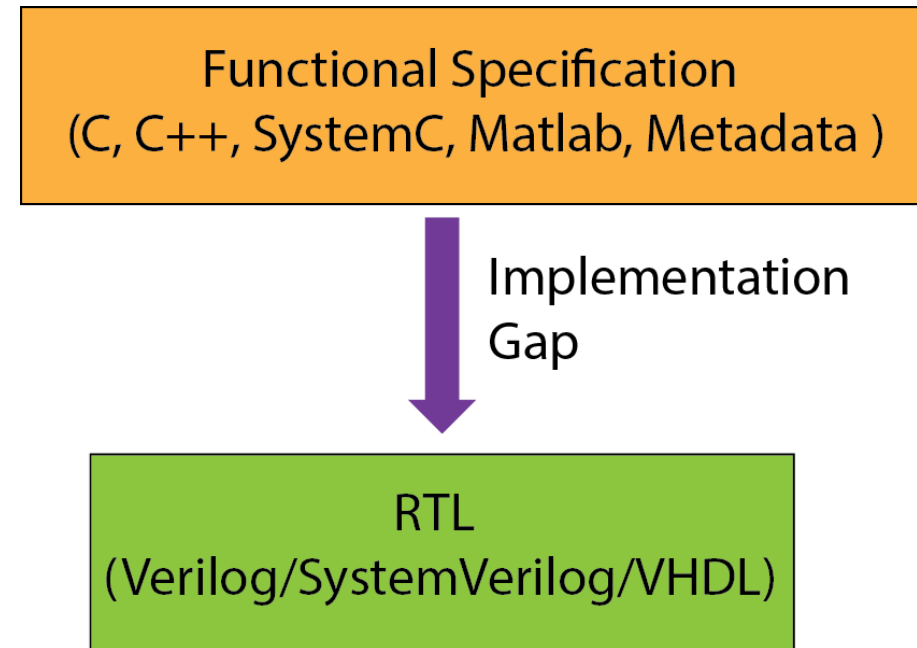


RTL: Register Transfer Level

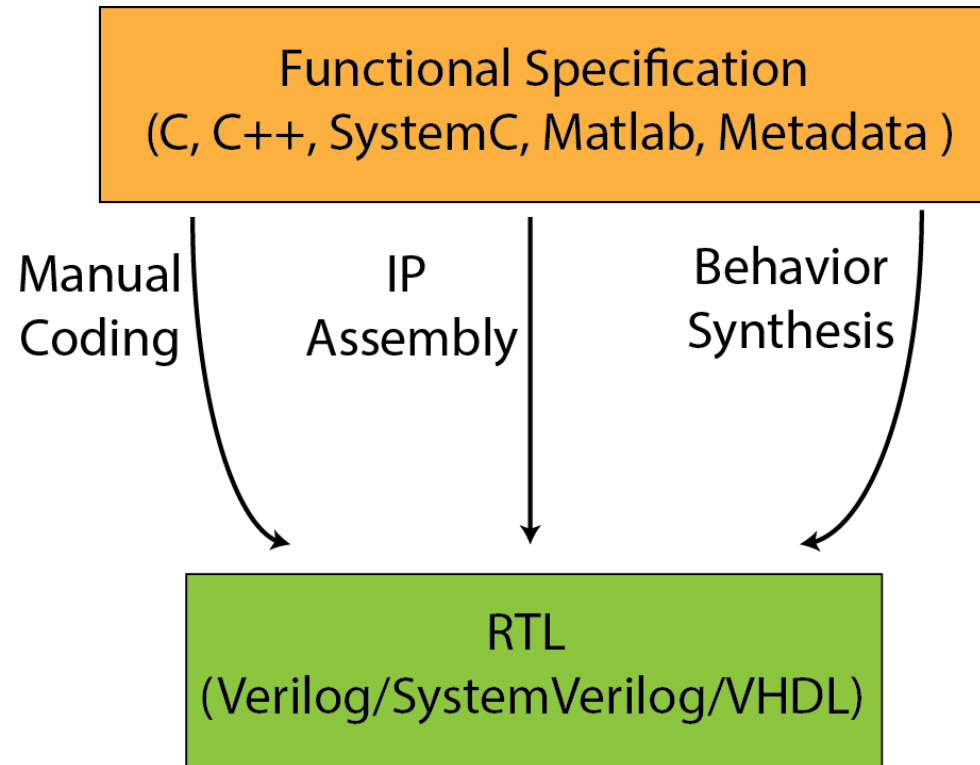
- Modelling of circuit as flow of data (signal) between *registers*
 - RTL can also be referred to as “data flow” description
-
- FSM generates control signals
 - MUX passes the data based on *control signals*
 - Computation is done on the data path

Functional Specification

- Functional Specification can be made at a higher level of abstraction
- Need to convert to RTL
 - Describes data flow from register to register at various time instants or clock cycle
 - Carries timing information
- Opens up *implementation gap*



Functional Specification to RTL

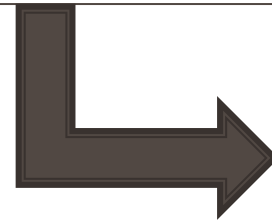


- Manual Coding: straight forward

- IP Assembly: reusing existing RTL

- Behaviour synthesis: automatic method of generating RTL from high-level language

Pre-RTL Methodologies



Reusing RTL

Functional Specification to RTL (1)

- Reusing RTL is especially popular in SoC (system-on-chip) design methodologies

System-on-chip (SoC)

- A complete system built on a single chip
- Composed of:
 - processors, hardware accelerators, memories, peripherals, analog components, and RF devices connected using some structured communication links
 - embedded software
- Improves productivity, lowers cost, increases features

Functional Specification to RTL (2)

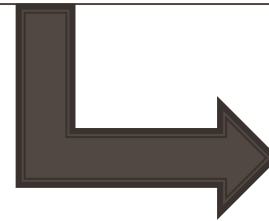
Intellectual Properties (IP)

- pre-designed and pre-verified sub-systems or blocks
- Example: processor, memory, interface, real-time operating system (RTOS)
- Sharing information

Integration of IPs (IP Assembly)

- Generator tools (using metadata)
- Configuration of IPs
- Communication links: network on chip (NoC)
- Verification Challenges

Pre-RTL Methodologies

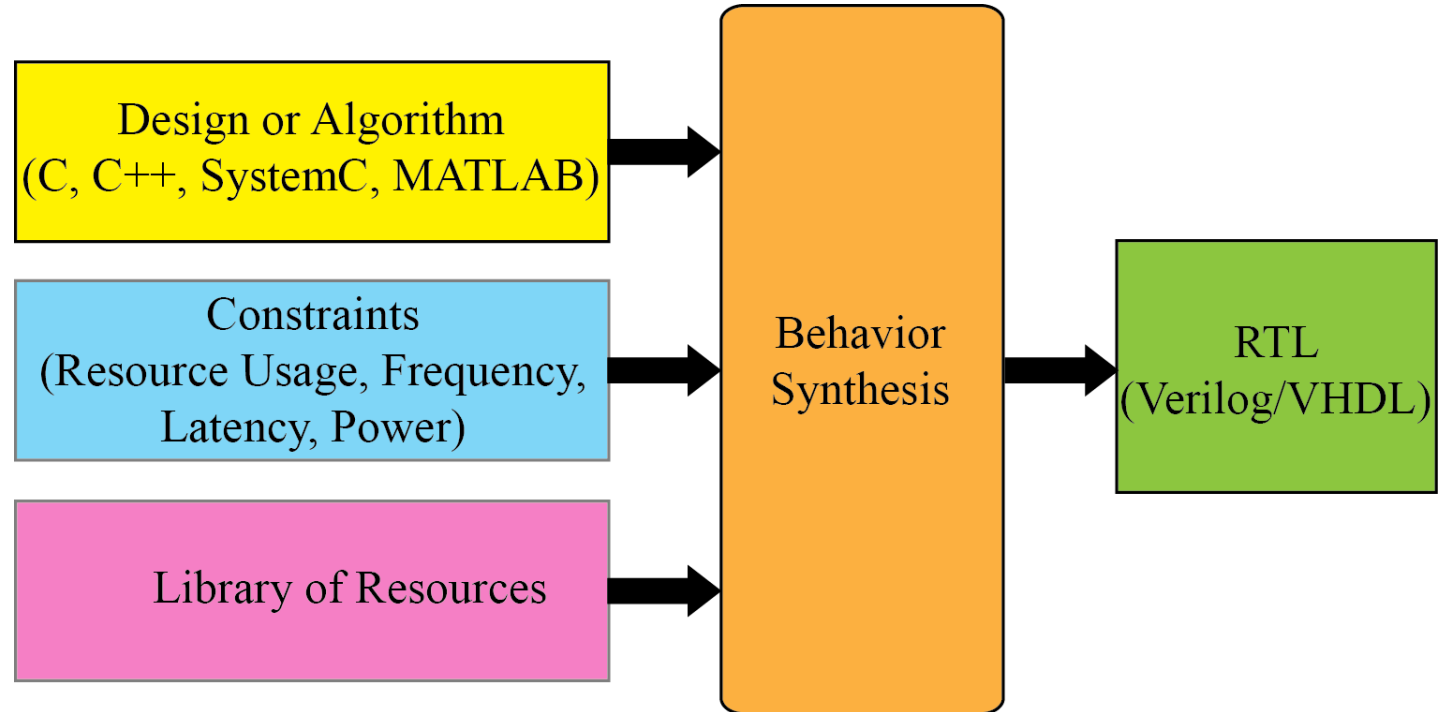


Behavior Synthesis

Behavioral Synthesis: What?

Behavioral Synthesis

- Process of converting a design that is described in algorithmic form that is not timed to an equivalent design in RTL that is fully timed and satisfy the constraints specified by the user.



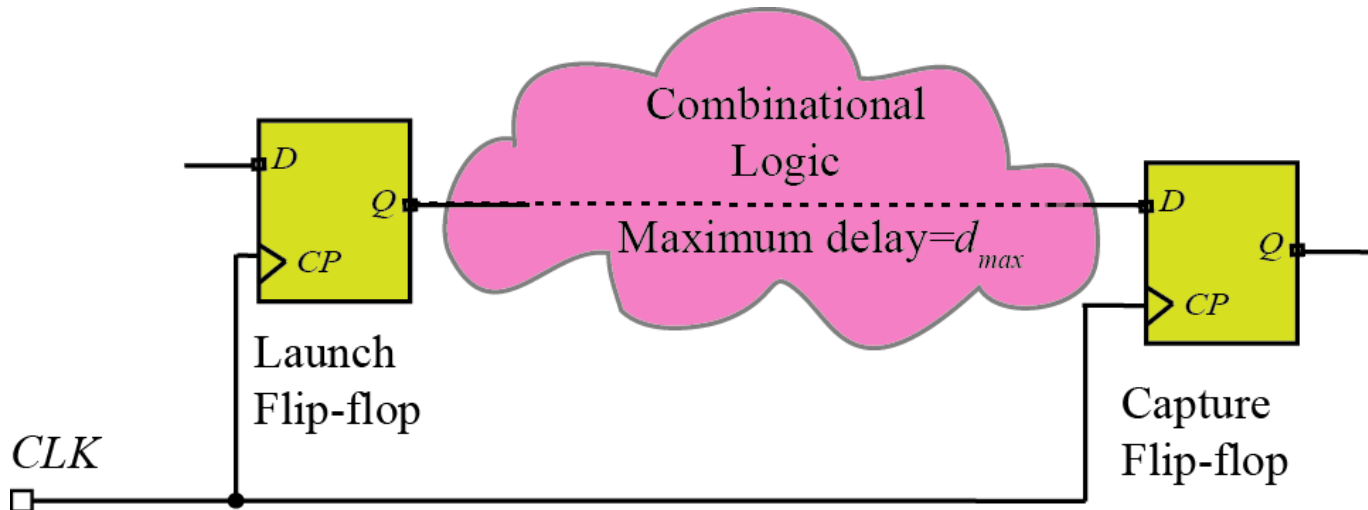
- Behavioral Synthesis is also called High-level Synthesis or Electronic System Level (ESL) synthesis

Behavioral Synthesis : Cost Metrics

- An untimed algorithm can be implemented in many different ways
- Different implementations can have different cost metrics

Cost Metrics:

- **Area:** number of circuit elements
- **Latency:** how many clock cycles are required before results are available
- **Worst case combinational delay**
- **Power dissipation, throughput, etc.**



- **Critical Path:** the combinational path that has the largest delay in the circuit (approximately)
- **Clock period** should be greater than the delay of the critical path ($T_p > d_{max}$)
- Maximum clock frequency $f_{max} < 1/d_{max}$

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.